

The Transformer: Self-Attention and Architecture

Scaler School of Technology | Deep Learning I | Week 7 — Session 1/2 | 08 May 2026



What Is This Lecture About?

In 2017, the “Attention Is All You Need” paper introduced the Transformer — an architecture that discarded RNNs entirely and relied solely on attention mechanisms. This lecture builds the Transformer from scratch, showing how self-attention, multi-head attention, and positional encoding combine to power GPT, BERT, and every modern large language model.

1 Self-Attention & Queries/Keys/Values

Every token simultaneously attends to every other token to determine relevance. We'll formalize this with Q, K, V matrices and scaled dot-product attention. *Analogy: a library search — your query matches book titles (keys) to retrieve content (values).*

2 Multi-Head Attention

A single attention head captures one type of relationship. Multiple heads run in parallel, each learning different patterns — grammar, coreference, semantics. *Analogy: multiple reviewers each checking a different aspect of your essay.*

3 Positional Encoding & Full Architecture

Attention is permutation-invariant — it needs explicit position information. We'll cover sinusoidal encoding, then assemble the complete encoder-decoder Transformer. *Analogy: numbering pages in a shuffled deck of cards.*

What to Expect in the Session

- Step-by-step attention computation walkthrough
- Visualization of attention weight heatmaps
- Interactive quiz on Q/K/V and scaling
- ★ Live demo: computing attention for a real sentence
- Comparison of Transformer vs. RNN complexity
- Preview of BERT and GPT for next session

How to Prepare (Optional but Helpful)

- ✓ **Brush up on** matrix multiplication and softmax sequences?
- ✓ **Recall** how Bahdanau attention works (Week 6)
- ✓ **Think about:** why do RNNs struggle with long sequences?
- ✓ **Skim** the “Attention Is All You Need” abstract (if curious)

Duration: ~2 hours **Prerequisites:** Sequence models, Bahdanau attention, matrix ops **Instructor:** Priyansh Saxena